Assignment1: House Price Prediction with Linear Models

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1. Introduction

This project predicts house prices using linear models, emphasizing strong performance through feature engineering rather than model complexity. Following the requirements, methods are limited to linear approaches (e.g., OLS and regularized models), with non-linear models used only as benchmarks. The goal is to balance predictive accuracy with interpretability and derive meaningful insights.

To support this goal, the analysis begins with a data preprocessing stage, including data cleaning and exploratory data analysis, to establish a reliable foundation for subsequent feature engineering and modeling.

2. Data Preprocessing

2.1 Data Cleaning

The raw dataset initially contained 9,200 observations with 18 features describing property characteristics and transaction information. A systematic data cleaning process was conducted to ensure data quality and reliability before modeling.

First, duplicate records were identified and removed. A total of 4,598 duplicated entries were detected, reducing the dataset to 4,602 unique observations. Removing duplicates was essential to prevent biased model estimation and over-representation of certain properties.

Second, data validity checks were performed on key variables. Several unrealistic price values were observed, including zero-valued transactions and extreme outliers exceeding 10 million. These values are unlikely to represent genuine market transactions and would distort model training. Therefore, observations with prices below 10,000 or above 10,000,000 were removed, resulting in a cleaner and more realistic price distribution.

Third, logical consistency checks were applied to structural features. Observations with zero bedrooms or zero bathrooms were identified and excluded, as these entries are not meaningful in a residential housing context. In addition, inconsistencies between living area and its components (e.g., mismatch between total living space and its sub-components) were detected and removed to ensure internal coherence of the dataset.

After cleaning, the final dataset contained 4,546 valid observations, providing a reliable foundation for subsequent analysis.

2.2 Target Variable Exploration

The distribution of house prices exhibited strong right skewness, with a long tail of high-value properties. To address this issue, a logarithmic transformation (base 10) was applied to the target variable. The log-transformed prices follow a much more symmetric and approximately normal distribution compared to the raw prices. This transformation is particularly beneficial for linear models, as it stabilizes variance and improves the linear relationship between predictors and the target.

3. Exploratory Data Analysis

3.1 Numerical Features

Analysis of numerical variables revealed significant right-skewness, particularly in lot size, necessitating log transformations to stabilize variance. Scatter plots confirmed a robust, approximately linear relationship between living area and log-transformed price, aligning with linear regression assumptions. In contrast, the relationship between lot size and price was weaker and more dispersed, suggesting that land size alone is not a strong predictor compared to living area.

In addition to univariate and bivariate analysis, a correlation heatmap was constructed to examine the relationships among numerical features. The heatmap reveals strong positive correlations between several size-related variables, particularly between sqft_living and sqft_above. This high correlation suggests the presence of multicollinearity, as these variables capture overlapping information about the usable living space. Since redundant information can inflate coefficient variance and impair model interpretability, these insights dictated a strategy of careful feature selection or consolidation in subsequent modeling stages. More broadly, the heatmap confirms that variables related to property size (e.g., living area, above-ground area, basement area) are among the most strongly correlated with price, reinforcing their importance as key predictors.

3.2 Categorical and Discrete Features

Boxplots were used to examine the relationship between discrete variables and house prices. The number of bedrooms shows a general upward trend with price, although the marginal effect diminishes at higher values. The number of floors also correlates with price, with multi-story houses generally priced higher. The “view” variable demonstrates a strong monotonic relationship with price, indicating that better views significantly increase property value. These findings suggest that both structural features and qualitative attributes contribute to price variation.

3.3 Temporal and Structural Insights

The distribution of construction years indicates that most properties were built after the 1950s, with a noticeable concentration in more recent decades. To further explore temporal effects, a new feature “decade” was created. Analysis of price per square foot across decades reveals that newer properties tend to command higher unit prices, suggesting a premium for modern construction.

The distribution of yr_renovated shows a highly skewed pattern, with a large proportion of properties having a value of zero, indicating no renovation. This observation suggests that treating renovation year as a continuous variable may not be appropriate. Instead, it is more meaningful to convert it into a binary indicator capturing whether a property has been renovated.

3.4 Location Effects

Location plays a critical role in house pricing. This was explored using both city-level and zip-code-level analysis. Boxplots of log-transformed prices across cities show substantial variation, indicating strong geographic heterogeneity. Further analysis at the zip-code level reveals that certain areas have significantly higher median prices. For example, the top zip codes exhibit median prices exceeding 2 million, highlighting the importance of spatial features in modeling.

These exploratory findings directly informed our feature engineering strategy: the skewed distributions motivated log transformations, the strong size-price relationship guided ratio-based features, the temporal patterns inspired lifecycle indicators, and the location heterogeneity called for sophisticated encoding strategies, which are detailed in Chapter 4.

4. Feature Engineering

Building on the insights obtained from the exploratory data analysis, the feature engineering process aims to translate observed data patterns into structured inputs suitable for linear models. Skewed distributions, size-related correlations, location heterogeneity, and temporal effects identified in the EDA guided the design of transformations, interaction terms, and aggregated features. To ensure methodological rigor and prevent data leakage, the dataset was first split into training (80%) and testing (20%) sets, and all distribution-dependent features were computed exclusively on the training set before being applied to the test data.

4.1 Core Feature Dimensions

The engineered features can be organized into four core dimensions that capture different aspects of property valuation.

Size and Spatial Structure. To address the right-skewed distributions observed in EDA, logarithmic transformations were applied to area-related variables, including `log_total_area` (total living space) and, as noted in Section 2.2, the target variable `log_price`. Ratio-based features were constructed to capture spatial efficiency: `basement_ratio` and `above_ratio` measure the composition of living space, while `living_lot_ratio` serves as a proxy for land utilization density. The `area_per_room` feature (living area divided by total rooms) distinguishes cramped multi-room houses from spacious layouts.

Property Lifecycle. House age at sale (house_age = sale_year - yr_built) captures temporal depreciation. To allow for nonlinear aging effects, a squared term (house_age²) was introduced, enabling the model to reflect the phenomenon where price declines may slow or reverse for older properties. Binary indicators were also created for new builds (new_house, age < 5 years) and historic properties (old_house, age > 40 years), allowing discrete intercept shifts for these market segments.

Quality and Premium Attributes. The living_condition interaction term (living area × condition) captures the amplified value of well-maintained large homes. View- and waterfront-related interactions—view_area (view × total area), waterfront_area (waterfront × total area), age_view_interaction (house_age × view), and luxury_view (view × waterfront)—encode the insight that scenic premiums scale with property size. Threshold-based indicators (large_house for area > 4,000 sqft, luxury_house for waterfront or high view scores) further isolate premium market tiers.

Structural Synergies. The bed_bath_interaction (bedrooms × bathrooms) captures the compounded utility of balanced room configurations, allowing the model to penalize properties with mismatched bedroom-to-bathroom ratios.

4.2 Location Encoding

Given the strong geographic heterogeneity observed in EDA, location information was encoded through multiple complementary strategies.

City-level Encoding. The city variable, with low cardinality, was transformed using one-hot encoding with the first category dropped to avoid perfect multicollinearity.

Zip-code Aggregates. For the high-cardinality statezip variable, target encoding was applied to avoid dimensionality explosion. Using only the training set, the mean price (zip_avg_price), standard deviation (zip_price_std), and observation count (zip_count) were computed for each zip code. These statistics were then mapped to both training and test sets, with unseen zip codes imputed using the global training mean. The log-transformed mean (log_zip_avg_price) was used in modeling.

Neighborhood Similarity Proxy. To capture micro-market dynamics beyond administrative boundaries, a K-Nearest Neighbors algorithm was deployed. Using sqft_living and sqft_lot as proximity coordinates, the five most similar properties within the training set were identified for each observation. The log-transformed average price of these neighbors (log_knn_avg_price) serves as a similarity-based proxy, mimicking the comparative market analysis approach used in real-world appraisal.

4.3 Final Feature Set

Table 1 summarizes the complete set of engineered features. All distribution-dependent statistics (target encoding aggregates and KNN averages) were derived strictly from the training partition to prevent data leakage.

Table 1: Summary of Engineered Features and Mathematical Formulations

Feature Name	Formula
log_price	ln(price)
total_area	sqft_above + sqft_basement
log_total_area	ln(max(total_area, 1))
area_per_room	sqft_living / (bedrooms + bathrooms + 1)
basement_ratio	sqft_basement / (sqft_living + 1)
above_ratio	sqft_above / (sqft_living + 1)
living_lot_ratio	sqft_living / (sqft_lot + 1)
house_age	sale_year - yr_built
house_age2	house_age * house_age
new_house	1 if house_age < 5 else 0
old_house	1 if house_age > 40 else 0
bed_bath_interaction	bedrooms * bathrooms
living_condition	sqft_living * condition
view_area	view * total_area
waterfront_area	waterfront * total_area
age_view_interaction	house_age*view
luxury_view	view * waterfront
large_house	1 if sqft_living > 4000 else 0
luxury_house	1 if (waterfront == 1) or (view >= 3) else 0
log_zip_avg_price	ln(mean price of statezip in training set)
zip_count	count of statezip in training set
zip_price_std	standard deviation of price for statezip in training set
log_knn_avg_price	ln(mean price of 5 nearest neighbors in training set based on sqft_living and sqft_lot)
city_XXX	1 if city == 'XXX' else 0 (drop_first=True)

5. Model Development and Evaluation

This section constructs and evaluates a system of predictive models for house prices, focusing on performance under comprehensive feature engineering. In addition to standard linear models (OLS, Ridge, Lasso), we introduced **ElasticNet** to balance L1 and L2 penalties and a **Multi-Spline Ridge** model to capture potential non-linearities in key numerical variables such as living area and house age. A XGBoost model is also trained as a nonlinear benchmark to contextualize the linear model results.

The modeling pipeline includes data preprocessing, train-test splitting, feature engineering (Chapter 4), model training, hyperparameter tuning via cross-validation, and evaluation based on Mean Absolute Percentage Error (MAPE). All models are

trained on log-transformed prices (log_price) to mitigate heteroscedasticity and reduce outlier influence; predictions are transformed back to the original price scale before computing MAPE.

5.1 Feature Set Overview

Building upon the feature engineering work detailed in Chapter 4, a total of 69 features were incorporated into the modeling pipeline, covering five key dimensions: property size, structure, quality, temporal factors, and location. Table 1 (Chapter 4) provides a complete summary of these features.

Regarding variable selection, this study adopts a "full inclusion + regularization-based selection" approach: all engineered features enter the model by default, with the shrinkage mechanisms of Ridge and Lasso regression automatically evaluating each variable's predictive contribution, rather than performing manual preselection. This approach preserves the integrity of the feature engineering work while allowing examination of different regularization methods in a high-dimensional feature space.

As shown in Figure 1, the KNN-based similarity feature (log_knn_avg_price) exhibits a clear linear association with the target variable, confirming that the "comparable sales method" translates effectively into a linear framework and provides valuable local market signals.

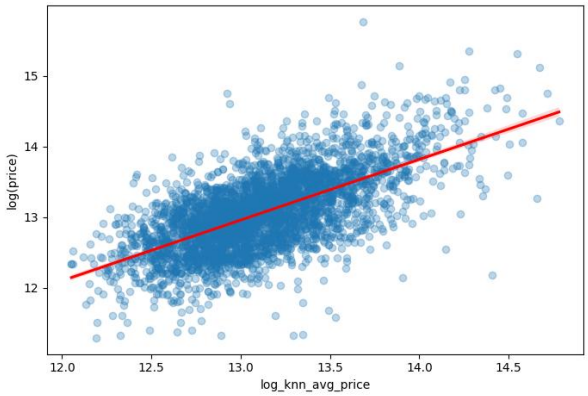

Figure 1. Relationship Between KNN Feature and Log Price

5.2 Model Training Setup

The dataset is split into 80% training and 20% testing to ensure objective evaluation. During training, 5-fold cross-validation is employed to tune parameters and prevent overfitting.

We implemented a multi-tiered modeling approach to evaluate performance:

- **OLS** serves as the baseline, minimizing the sum of squared errors.
- **Ridge** applies L2 regularization ($\alpha = 10$) to mitigate multicollinearity.
- **Lasso** applies L1 regularization ($\alpha = 0.001$), enabling feature selection through coefficient shrinkage.
- **ElasticNet** balances L1 and L2 penalties for high-dimensional data with

complex correlations.

- **Multi-Spline Ridge** incorporates B-splines (df=6) to capture local non-linear trends within a linear framework.

5.3 Evaluation Metric

Model performance is evaluated using Mean Absolute Percentage Error (MAPE):

$$\text{MAPE}=\frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i-\hat{y}_i}{y_i}\right|$$

This metric measures relative prediction error, making it well-suited for house price data with large scale variations. All predictions are transformed back to the original price scale before computing MAPE. The average MAPE across 5-fold cross-validation is also reported to assess stability.

5.4 Model Performance Assessment

5.4.1 Linear Model Comparison

Table 2 summarizes the performance of linear models and XGBoost on the test set.

Table 2. Model Performance Comparison

Model	Train MAPE	Test MAPE
OLS	0.1625	0.1600
Ridge	0.1639	0.1606
Lasso	0.1682	0.1663
ElasticNet	0.1644	0.1614
Multi-Spline	0.1623	0.1606
XGBoost	0.1097	0.1564

OLS model achieved a test MAPE of 0.1600, emerging as the top-performing linear estimator. This underscores the effectiveness of our feature engineering pipeline, which successfully captured primary price determinants without introducing overfitting, even in the absence of explicit regularization.

Ridge and Multi-Spline Ridge models yielded nearly identical results (MAPE ≈ 0.1606). This marginal performance delta suggests that while L2 regularization and B-splines offer theoretical robustness against multicollinearity and non-linearity, the log-transformed feature set already encapsulated the majority of the predictive signal.
In contrast, Lasso (0.1663) and ElasticNet (0.1614) exhibited higher error rates. Lasso's performance drop indicates that its sparsity constraint may have excluded features with meaningful marginal value.

The high consistency between CV and test results across all architectures indicates strong generalization. Furthermore, the negligible margin between OLS (0.1600) and the non-linear XGBoost benchmark (0.1564) reinforces the conclusion that our feature engineering effectively linearized dominant market patterns.

Figure 2 plots predicted versus actual prices for the Ridge model. Most predictions

align closely along the 45-degree reference line, although deviations increase for high-value properties—a pattern examined further in Section 5.4.3.

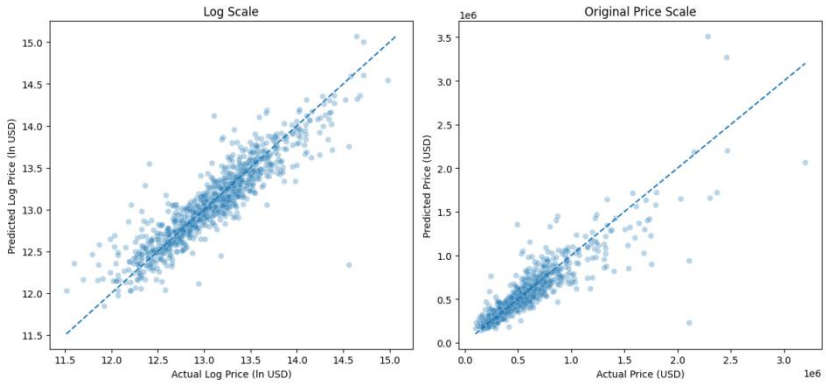

Figure 2. Actual vs Predicted Prices (Ridge Model)

5.4.2 Nonlinear Benchmark: XGBoost

To provide a performance upper bound, an XGBoost model was trained on the same feature set. With optimized hyperparameters (n_estimators = 700, max_depth = 5, learning_rate = 0.03, reg_alpha = 1, reg_lambda = 5), the model achieved a test MAPE of **0.1564**, nearly identical to the best linear model's **0.1600** — a gap of just 0.0036. Notably, the default XGBoost (Train MAPE = 0.0587, Test MAPE = 0.1597) suffers severe overfitting, underscoring the importance of proper regularisation. The narrow margin between the tuned XGBoost and the linear models confirms that the feature engineering pipeline successfully captured the dominant sources of price variation within a linear framework.

5.4.3 Residual Diagnostics

To further evaluate model fit and identify systematic biases, residual analysis was conducted for the Ridge model. Figure 3 plots residuals against predicted log prices, with a LOWESS smooth curve to highlight patterns.

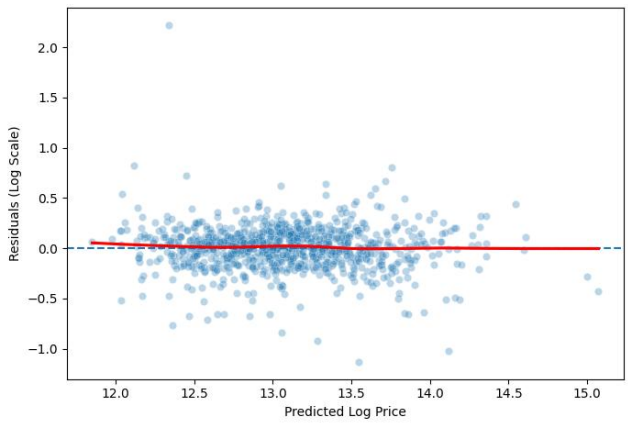

Figure 3. Residual Plot of Predicted Log Price (Ridge Model)

The LOWESS curve shows an upward deviation at the lower price range, indicating systematic underestimation of low-priced properties, likely due to higher heterogeneity and unobserved factors not captured by the model. In contrast, the curve remains flat for mid-to-high prices, suggesting minimal bias. The lack of curvature at

higher prices reflects the strong anchoring effect of location and comparable-sale features (e.g., log_zip_avg_price and log_knn_avg_price), which support accurate predictions in mainstream and premium segments.

5.4.4 Performance Summary

The model performance assessment reveals that all linear models—including OLS, Ridge, Lasso, ElasticNet, and Spline-enhanced Ridge—achieved highly consistent test MAPEs ranging from **0.1600 to 0.1663**. The performance gap between the best linear model and the XGBoost benchmark (MAPE **0.1564**) is a modest **2.25%**, confirming that our feature engineering effectively captured the vast majority of non-linear patterns within a linear framework. While residual diagnostics indicate a systematic underestimation of low-priced properties due to localized volatility, the high consistency between cross-validation and test results proves that the models generalize well without overfitting. These findings underscore that the linear models provide a compelling balance of simplicity and accuracy, yielding results remarkably close to complex ensemble methods while maintaining full economic interpretability.

6. Model Interpretability

To gain deeper insights into the drivers of property valuation, this study analyzes feature importance based on the standardized coefficients of the Ridge model. Compared to OLS, Ridge regression provides more stable coefficient estimates in the presence of multicollinearity, making it more suitable for interpretability and structural economic analysis. The final model integrates multiple dimensions, including property size, structural design, geographic location, housing quality, and temporal effects.

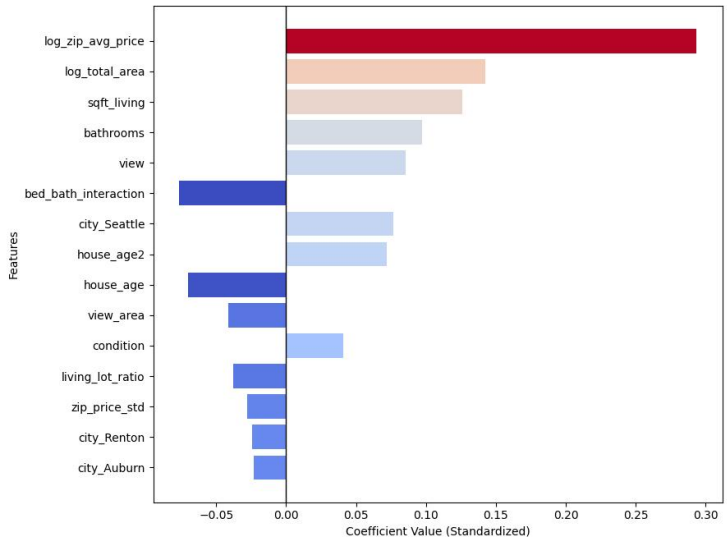

Figure 4. Top 15 Ridge Feature Coefficients

6.1 Dominant Factors: Location and Property Scale

Location is the strongest determinant of house prices. The most influential variable, log_zip_avg_price (+0.294), far exceeds all others, indicating that neighborhood-level price conditions strongly anchor individual property values. Property scale is the

second key factor. log_total_area (+0.142) and sqft_living (+0.126) both show strong positive effects, confirming that larger homes are consistently more valuable. bathrooms (+0.097) further highlights the importance of functional living space, indicating that usability and livable capacity are key drivers of housing demand.

6.2 Structural Features and Temporal Dynamics

From a structural perspective, **bed_bath_interaction (-0.077)** shows a negative coefficient, suggesting diminishing marginal utility when the balance between bedrooms and bathrooms is suboptimal. This reflects buyer preferences for functional coherence rather than simple additive room counts.

In terms of temporal effects, **house_age (-0.070)** is negatively associated with price, confirming depreciation with aging properties. However, **house_age2 (+0.072)** indicates a nonlinear (convex) relationship, implying that depreciation slows over time. In some cases, older properties may stabilize in value or even regain relative attractiveness due to scarcity effects or historical character.

6.3 Geographic Effects and Quality Premiums

City-level variables reveal clear spatial heterogeneity in housing prices. **city_Seattle (+0.076)** exhibits a significant urban premium, reflecting higher baseline prices in core metropolitan areas. In contrast, suburban locations such as **city_Renton (-0.024)** and **city_Auburn (-0.023)** are associated with price discounts relative to the baseline, highlighting a pronounced spatial hierarchy within the housing market. Quality features also matter: **view (+0.085)** and **condition (+0.041)** both increase prices. In contrast, **view_area (-0.041)** suggests diminishing returns or overlap with other size-related variables.

6.4 Key Takeaways for Stakeholders

Overall, housing prices are driven by a hierarchical combination of factors rather than a simple additive structure:

- **Location dominates all factors**, especially neighborhood price level.
- **Property size and functional capacity are key drivers**, particularly total area and bathrooms.
- **Structural efficiency influences value**, not just absolute size.
- **Age effects are nonlinear**, with depreciation slowing over time and potential stabilization in older homes.
- **Clear spatial stratification exists**, with Seattle commanding a premium relative to suburban areas such as Renton and Auburn.

These findings demonstrate that, with effective feature engineering, Ridge regression not only delivers strong predictive performance but also provides a transparent and economically interpretable framework for real estate valuation and decision-making.

AI Statement: We used **AI tools** to assist us in completing this task.